

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (E&C) (Credit System) Degree Examinations
May - June 2023

20ECE45 - INTERNET OF THINGS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain with a neat sketch, the framework of Internet of Things.	10	L*2	1	1
b)	Describe the layer 'Technology' and explain the three layers in the high-level view of an IoT architecture.	10	L2	1	1
2. a)	Explain the role of sensor nodes in WSN (Wireless Sensor Networks).	6	L2	2	1
b)	Why actuators are called as 'Effectors'?	4	L2	2	1
c)	What is RFID and explain how does it work?	10	L2	1	1
3. a)	Compare and contrast the following:				
i)	IPv4 Vs IPv6	5+5	L2	1	1
ii)	TCP Vs UDP				
b)	Illustrate the working of HTTP protocol in IoT applications.	5	L2	2	1
c)	Justify the statement "CoAP maps to HTTP".	5	L2	2	1
Unit – II					
4. a)	Explain Cloud-based solution. What are its disadvantages?	5	L2	3	1
b)	List out and explain any two advantages of Pub/Sub model.	5	L2	3	1
c)	Compare and contrast MQTT and AMQP.	10	L3	3	1
5. a)	Briefly explain UPnP, SLP and Zero-conf and how they are related to each other.	8	L2	3	1
b)	Define Cloud computing in IoT.	4	L2	3	1
c)	Explain fog computing and mention its characteristics.	8	L2	4	1
6. a)	Explain the 'multi-V' model in Big Data.	5	L2	4	1
b)	Compare Big data with Big Stream in IoT scenario.	5	L2	3	1
c)	What is meant by 'Virtualization' of IoT hubs and what are its benefits?	10	L3	3	1
Unit – III					
7. a)	Write a program to detect the motion with PIR sensor using Raspberry Pi.	10	L2	5	1
b)	Explain the major features of RPi hardware platforms. (Min. 5 points).	10	L2	5	1
8. a)	Explain in detail with a neat sketch, the high-level view of the main hardware components in a smart object with wired communication interface.	10	L3	5	1
b)	Write an application code allowing the user to control the state of the LED through a simple browser.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
